

Fig. 3: Shaft setting

cuts the internal light beam within the slot of the optocoupler (IC2). As a result, pulses are generated. This train of pulses is applied to input pin 1 of AND gate N1. Pin 2 of AND gate N1 is connected to

output pin 3 of IC1.

To measure the RPS of the shaft, press switch S2 momentarily to enable gate N1 for time duration of one second via the monostable multivibrator (IC1). The pulses are applied as clocks to IC5. The counter increments and generates the code to drive the seven-segment displays. The displays show the number of counts in one second. Accordingly,

you can simply calculate revolutions per minute (RPM).

Assemble the circuit on a general-purpose PCB and enclose in a cabinet with enough space for the battery, switches and motor. Connect the 9V battery and the module is now ready to display the RPS of the motor. As shown in Fig. 3, fix a nose on the motor shaft so that it can cut the internal IR beam of the optocoupler. ●

INFRARED ILLUMINATOR

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Infrared (IR) illuminators are widely used to improve the image-capturing quality of security cameras fitted in dark zones. Just like our eyes, cameras also can't record movements

although when these LEDs are used for security applications, some individuals can detect a very dim red glow from them. The 940nm LED radiations are completely invisible to the eye. Some of these LEDs are clear, while others are tinted with pale shades of

needed.

The circuit (refer Fig. 1) can be divided into three parts: ambient light sensor, relay driver and IR LEDs. The ambient light sensor is built around

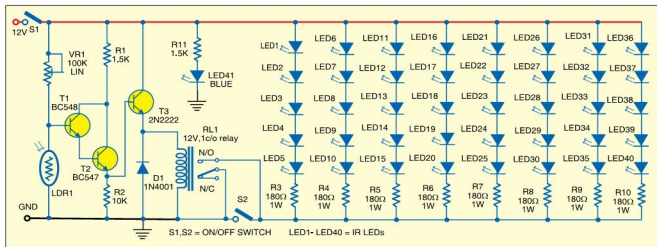


Fig. 1: Circuit for infrared illuminator

in dark. However, unlike our eyes, most of the latest cameras can capture infrared light.

In an IR Illuminator, many infrared IR LEDs are grouped together to throw good amount of IR light. Typically, LEDs output at 470 nm (blue region), 525 nm (green region) and 625 nm (red region). IR LEDs produce longer wavelengths, 880 nm and 940 nm being the common ones. Most CCD cameras are a little more sensitive to 880 nm,

dark gray, blue or even black. They come in various configurations and radiation patterns, but 5mm types with 15- to 40-degree patterns are the most popular.

Typically, IR LEDs run at around 1.3 to 1.7 volts, depending on the LED current (typically 10 to 30 mA). However, this may vary with the type and manufacturer. Practically, IR illuminators may have 6 or 60 to 100 or more LEDs, depending on the output

multiturn linear potmeter VR1 and light-dependent resistor LDR1. The relay driver section is built around transistors T1 through T3. The IR LEDs section is built around LED1 through LED40.

The light sensor circuit is a simple transistor switch with the base of the Darlington pair (formed by T1 and T2) connected to a voltage divider. Variable resistor VR1 and the 10mm encapsulated LDR are used to sense the

